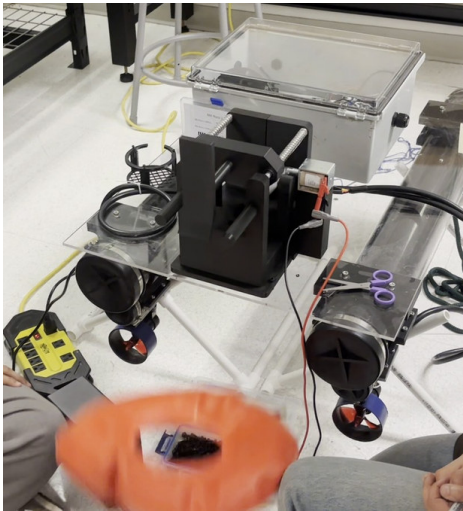


DROWNING PREVENTION ROBOT

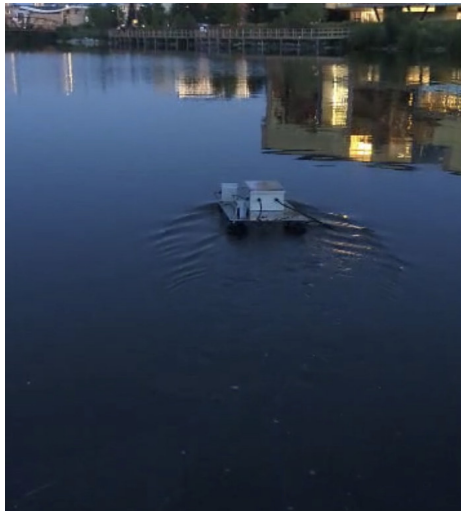
SOFTWARE TEAM LEAD | 2026 MASSROBOTICS FORM & FUNCTION FINALIST



UW MedTech Robotics



FULL ROBOT / BENCH INTEGRATION



AUTONOMOUS WATER TEST



ETHERNET LIDAR INTEGRATION

What?

- Build a sealed autonomous surface robot for drowning prevention as part of the UW MedTech Robotics design team.
- Owned autonomy, embedded integration, sensor software, and the later ROS 2 architecture.

How?

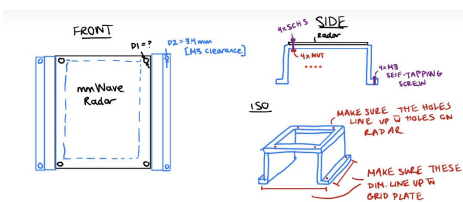
- Integrated **GNSS over UART**, **BNO055 over I2C**, USB camera, **Ethernet LiDAR**, and an ESP32 controlling dual thrusters.
- Built browser-based manual control plus map-selected waypoint autonomy; rewrote the monolith as isolated **ROS 2 nodes**.
- Replaced the 6-axis MPU6050 with a 9-axis BNO055 for calibrated absolute heading; stale data and serial timeouts return thrust to neutral.

Results

- Water-tested manual and autonomous operation using GNSS, absolute heading, and differential thrust.
- Completed live camera streaming and high-resolution **Ethernet LiDAR integration**.
- Selected as a **2026 MassRobotics Form & Function finalist**.

RADAR PERCEPTION + WATERPROOF MOUNT

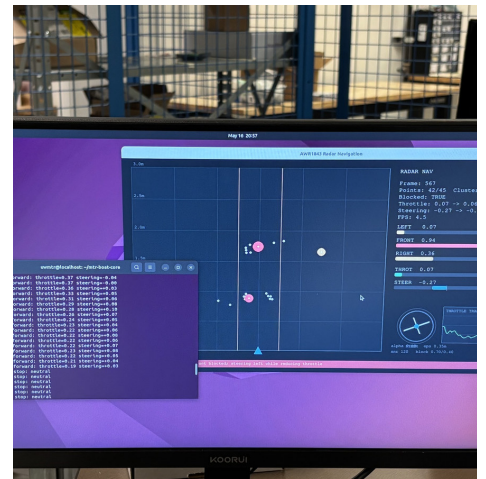
DPR SUBSYSTEM | TI MMWAVE | PYTHON | CAD + TESTING



FULL MOUNT SKETCH



PRINTED MOUNT IN ENCLOSURE



CLUSTERED OBSTACLE SECTORS

Problem

- Raw **binary UART frames** contained noisy reflections; enclosure spacing also reduced detected points.
- At **3.5 cm** behind the clear face, the radar retained only **34% of points**.

Development

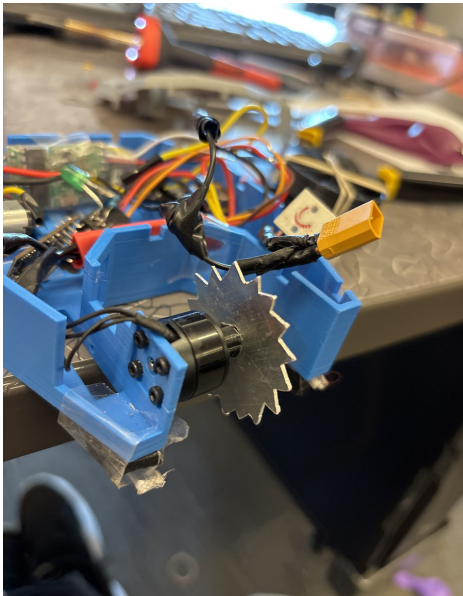
- Decoded point clouds and tested **SNR** and **spatial filters**.
- Grouped detections with **density-based clustering**; scored confidence using count, distance, and signal quality.
- Designed a mount that fixed the sensor **0.2 cm** from the waterproof window.

Result

- Point retention increased from **34% to 81%**.
- Temporal smoothing and **hysteresis** stabilized left/front/right obstacle evidence.
- Converted clusters into steering, throttle, and stop commands in small-scale navigation tests.

150 G COMBAT ROBOT - DREADBOT

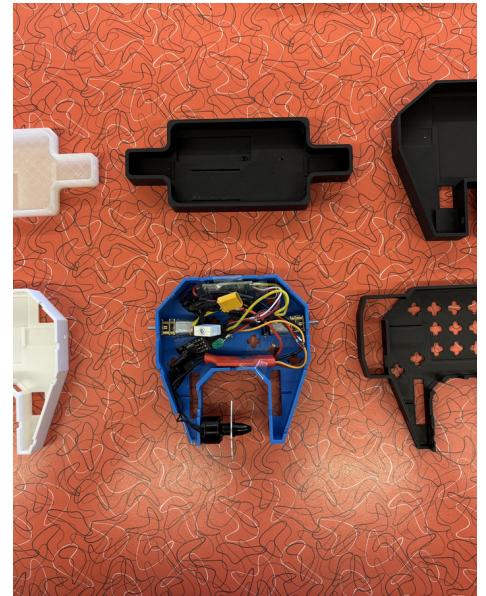
SOLIDWORKS | ADDITIVE MANUFACTURING | FABRICATION



COMPETITION ROBOT



HAND-FABRICATED ALUMINUM WEAPON



PHYSICAL ITERATIONS

What?

- Design and build a **150 g** vertical-spinner robot for the UW student Battlebots competition.
- Completed **100%** of the mechanical CAD and packaging in a two-week concept-to-arena sprint.

How?

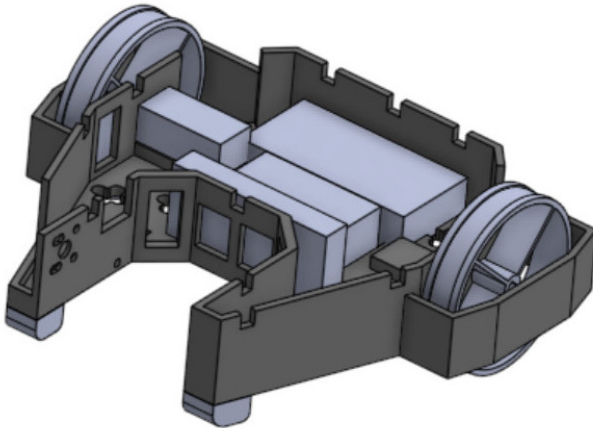
- Balanced wheel clearance, weapon support, electronics access, and mass inside a **3D-printed PLA chassis**.
- Hand-fabricated the aluminum spinner by drilling its center bore and filing the tooth profile.

Results

- Two final prints failed on competition morning; retaining the previous chassis revision kept the robot operational.
- **One of 7 functioning robots among 20+ entries**; competition exposed clear mechanical and electrical failure modes.

FAILURE ANALYSIS + RAGE REDESIGN

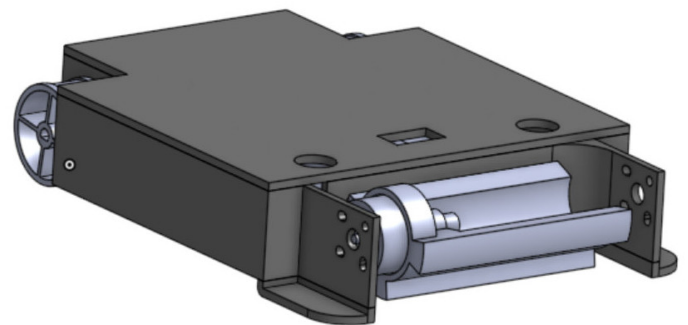
DREADBOT LESSONS APPLIED TO THE NEXT COMPETITION DESIGN



DREADBOT CAD / COMPETITION DESIGN

Observed failure

- Large wheels enabled inverted driving but raised the chassis and made impacts destabilizing.
- The thin aluminum saw deformed and carried too little **rotational inertia** for strong energy transfer.
- Rushed wiring contributed to weapon-motor failure after impacts.



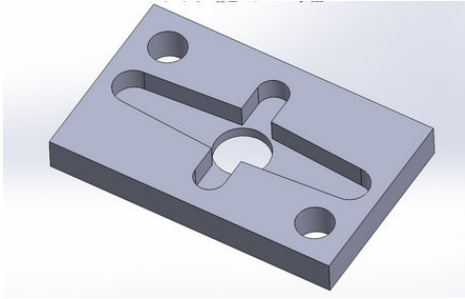
RAGE CAD / IN DEVELOPMENT

Rage response

- Lower center of gravity and protected wheels; invertibility is deliberately sacrificed for survival.
- Wider, heavier **high-inertia** impact weapon plus more deliberate electronics packaging and wiring.
- Current status: **CAD complete**; fabrication and validation planned for the next competition cycle.

HEAD-TRACKING TURRET

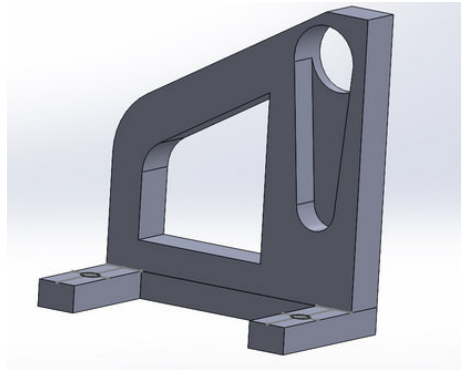
SOLIDWORKS | 3D PRINTING | ARDUINO | OPENCV



BASEPLATE CAD / V5

What?

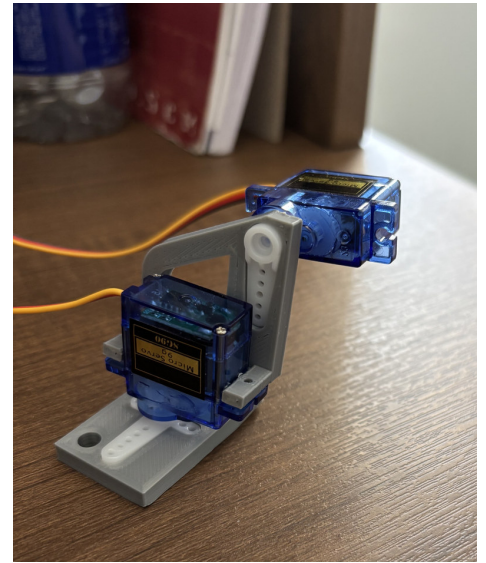
- Designed and built a two-axis turret that detects and follows a human face in real time.
- Translated webcam detections into physical pan and tilt motion.



TILT MECHANISM CAD / V2

How?

- Modeled the baseplate and tilt bracket in **SolidWorks**, designed clearance fits, then 3D printed and assembled the mechanism.
- Sent **OpenCV face coordinates** over serial to an Arduino controlling two servo motors.



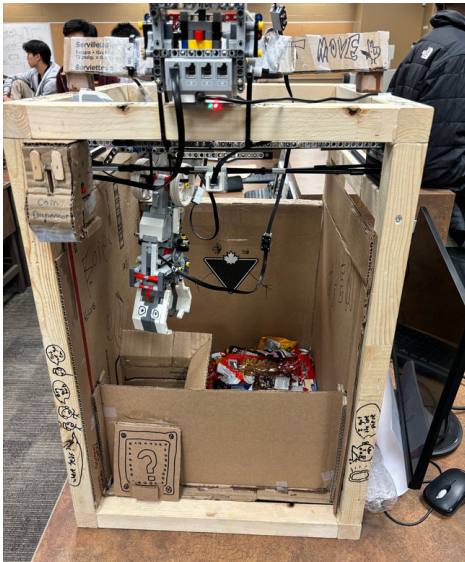
PRINTED PAN-TILT ASSEMBLY

Result

- The assembled turret accurately followed face movement through both axes.
- Validated a complete CAD-to-print workflow alongside embedded control and computer vision.

CLAW MACHINE

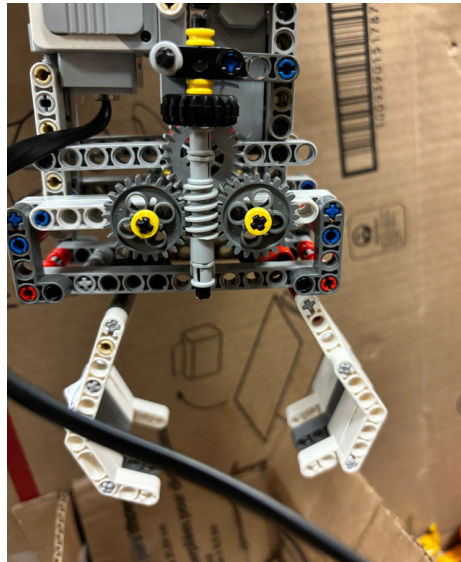
MECHANISM CO-DESIGN | ROBOTC | LEGO EV3



COIN-OPERATED MACHINE

What?

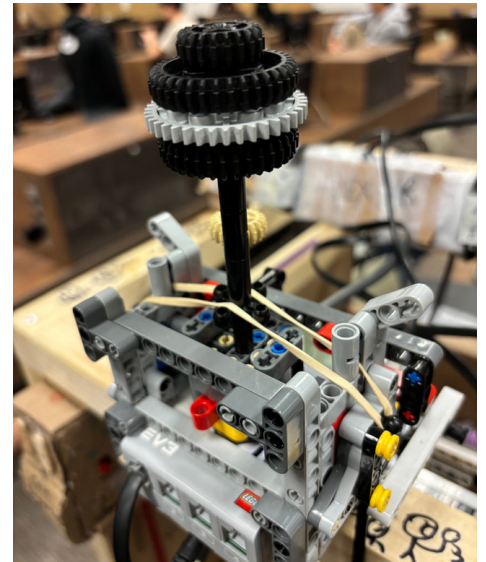
- Team of four built a functional coin-operated claw machine using custom mechanics and a **LEGO EV3 controller**.
- Served as **sole RobotC developer** and co-designed the claw mechanism.



WORM-GEAR CLAW

How?

- Programmed the **XYZ gantry**, claw sequence, joystick interface, game states, and on-screen display.
- Co-developed the **LEGO Technic worm-gear claw** for gripping strength and repeatable motion.



CUSTOM JOYSTICK

Result

- Delivered a fully operational machine integrating structure, gearing, electronics, and software.
- Attracted **more than 50 users** and generated **\$5.05** during a University of Waterloo showcase.